

Reading List — σ -Essential Problem

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Personal reference

June 2026

The object: a concrete σ -complete non-Boolean irreducible OML carrying a finite two-valued state on a $\perp$ -closed subposet that extends to no global σ -additive two-valued state. Localized to σ -point-selection failure; frontier is non-Polish / non-standard-Borel / irreducibly non-central. Sources in dependency order, each with the question to read it for.

1 Descriptive set theory

Kechris, *Classical Descriptive Set Theory* (GTM 156). Borel / analytic / co-analytic hierarchy $\rightarrow$ Borel equivalence relations $\rightarrow$ the Glimm–Effros dichotomy; then uniformization.

- State Glimm–Effros precisely (non-smooth $\iff E_0$ embeds $\iff$ no Borel two-valued transversal). Which word does not transport to clause (ii)? (“Borel.”)
- Co-analytic uniformization-failure: why is its obstruction real-valued / over a standard-Borel base, hence Derr–Williamson-killed?

Gao, *Invariant Descriptive Set Theory* (after Kechris). E_0 / E_∞ / hyperfinite / turbulence.

- Define hyperfinite; state Adams–Kechris. Why is E_∞ non-hyperfiniteness orthogonal to clause (ii)?

2 Large cardinals and forcing

Jech, *Set Theory* (3rd millennium ed.).

- Filters, ideals, ultrafilters: why is $\mathcal{F}_s = \{A : s(A) = 1\}$ not a filter on a non-Boolean concrete OML, even for a Dirac state? (MO_2 : $a \cap b \neq \emptyset$ as sets yet $a \wedge b = 0$.)
- Measurable cardinals, Ulam matrices: re-derive the matrix by hand; locate the line needing non-orthogonal unions / comprehension.
- Forcing: why does set-forcing preserve measurability (Lévy–Solovay)?

Kanamori, *The Higher Infinite* (after Jech, if the strength path calls). Measurable $\rightarrow$ ultrapower $\rightarrow$ elementary embedding $j: V \rightarrow M$.

- The UB route transports U along j , but U lives on Boolean $\mathcal{P}(\kappa)$; why is transport to orthogonal-only a costume?

3 Primary sources (read for technique)

- **Navara–Pták 1983.** Re-derive their ZFC non-Dirac σ -state. What makes $L_{f,g}$ support it, and why is it not a witness?
- **Derr–Williamson 2023 + Maharam 1972 §8.** Where is inner-regularity load-bearing in Thm D.6 / Maharam §8.2?
- **Blecher–Weaver 2017.** Identify the ultrafilter-from-masa step (Prop. 2.3); state the masa-free question it leaves open for the concrete two-valued case.

4 Notes

- Start: Kechris (Borel equivalence relations $\rightarrow$ Glimm–Effros).
- Definability-vs-existence first: bounds §3f and §3j, plus the Lean polarity gate.
- Operator algebra (masas, von Neumann algebras) is not on the critical path: the masa route is dead, so light functional analysis suffices for Blecher–Weaver.
- Chapter numbers are convenience pointers; navigate by topic.